Program: **Diploma in Microelectronics**	
Course Code: **5493D**	Course Title: **VLSI Testing**
Semester: **5**	Credits :**4**
Course Category: **Program Elective course**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- to introduce the VLSI testing
- to introduce logic and fault simulation and testability measures
- to study the test generation for combinational and sequential circuits
- to study the design for testability.
- to study the fault diagnosis

Course Prerequisites:

Topic	Course code	Course name	Semester
Combinational Circuits		Digital Electronics	3
Sequential Circuits and its types		Digital Electronics	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Understand VLSI Testing Process	11	Understanding
CO2	Develop Logic Simulation and Fault Simulation	20	Understanding
CO3	Develop Test for Combinational and Sequential Circuits	14	Understanding
CO4	Understand the Design for Testability	13	Understanding
	Series Tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2					
CO3	3	3	3				
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand VLSI Testing Process**		
M1.01	Explain VLSI Testing Process	3	Understanding
M1.02	Explain VLSI Testing Equipment	3	Understanding
M1.03	Explain fault models	3	Understanding
M1.04	Describe the relationship Among Fault Models.	2	Understanding
Contents: **INTRODUCTION TO TESTING:** Introduction – VLSI Testing Process and Test Equipment – Challenges in VLSI Testing - Test Economics and Product Quality – Fault Modeling – Relationship Among Fault Models.			
CO2	**Identify and Develop fault models in logic circuit**		
M2.01	Explain Simulation for Design Verification and Test Evaluation	3	Understanding
M2.02	Explain Modeling Circuits for Simulation	4	Understanding
M2.03	Explain the algorithms for true value and fault simulation	6	Understanding
M2.04	Find the faults in the given logic circuit	7	Applying

	Series Test 1	1	

Contents :

LOGIC & FAULT SIMULATION & TESTABILITY MEASURES: Simulation for Design Verification and Test Evaluation – Modeling Circuits for Simulation – Algorithms for True Value and Fault Simulation

Fault Modelling: Boolean Difference Method, Path sensitization method, Kohavi Algorithm

CO3	Develop Test for Combinational and Sequential Circuits		
M3.01	Explain Combinational ATPG Algorithms	3	Understanding
M3.02	Explain Sequential ATPG Algorithms	3	Understanding
M3.03	Explain Simulation based ATPG	4	Understanding
M3.04	Explain Genetic Algorithm based ATPG	4	Understanding

Contents:

TEST GENERATION FOR COMBINATIONAL AND SEQUENTIAL CIRCUITS Algorithms and Representations – Redundancy Identification – Combinational ATPG Algorithms – Sequential ATPG Algorithms – Simulation Based ATPG – Genetic Algorithm Based ATPG

CO4	Understand the Design for Testability		
M4.01	Explain Testabilty basics and its analysis	4	Understanding
M4.02	Explain BIST architecture	4	Understanding
M4.03	Explain Fault Models for Diagnosis	5	Understanding
	Series Test 2	1	

Contents:

DESIGN FOR TESTABILITY: Design for Testability Basics – Testability Analysis - Scan Cell Designs – Scan Architecture – Built in Self-Test – Random Logic Bist

FAULT DIAGNOSIS: Introduction and Basic Definitions – Fault Models for Diagnosis – Generation of Vectors for Diagnosis – Combinational Logic Diagnosis - Scan Chain Diagnosis – Logic BIST Diagnosis.

Text/Reference:

T/R	Book Title/Author
T1	Laung-Terng Wang, Cheng-Wen Wu and Xiaoqing Wen, “VLSI Test Principles and Architectures”, Elsevier, 2017
T2	Michael L. Bushnell and Vishwani D. Agrawal, “Essentials of Electronic Testing for Digital, Memory & Mixed-Signal VLSI Circuits” , Kluwer Academic Publishers, 2017.
R1	Niraj K. Jha and Sandeep Gupta, “Testing of Digital Systems”, Cambridge University Press, 2017.
R2	Parag K. Lala, “An Introduction to Logic Circuit Testing,” Morgan Claypool Publishers, 2009.
R3	Parag K. Lala, “Digital Circuit Testing and Testability,” Academic Press, 1997.

Online Resources:

Sl. No.	Website Link
1	https://onlinecourses.nptel.ac.in/noc21_ee39
2	https://onlinecourses.nptel.ac.in/noc25_ee24/preview
3	https://www.geeksforgeeks.org/principles-in-digital-system-design/